NewSpace Economy Start-up Written Report

Section 1: Technical Feasibility and Physical Advantages

Classical silicon architectures face insurmountable thermal and lithographic limits on Earth. By relocating Quantum Processing Units (QPU) fabrication to orbit, we leverage microgravity and near-infinite vacuum to bypass terrestrial decoherence and structural sag. Terrestrial quantum architectures, particularly trapped-ion and neutral-atom systems, expend significant energy and hardware overhead using magnetic/optical tweezers to counteract Earth's 1g environment. In microgravity, ions remain stationary, allowing magnetic resources to be reallocated entirely to high-fidelity gate operations.

While Earth-bound cryostats struggle to maintain 10^{-10} Torr, the orbital environment provides a high-quality vacuum naturally in the wake of the craft. This minimizes collisions between qubits and background gas molecules, which is a primary driver of decoherence. However, space is a radiative environment rather than a convective one; therefore, we utilize Active Magnetic Regenerative Refrigeration (AMRR) and oversized radiator panels to bypass the need for gravity-dependent helium isotopes, turning space into a functional heat sink for the 20 mK operating temperatures required.

3D Nanoprinting and In-Situ Integration

Our core innovation is the 3D nanoprinting of qubit arrays directly in orbit. This process eliminates the "sag" and thermal-expansion distortions inherent in Earth-based lithography, enabling the creation of larger, more uniform qubit lattices. We utilize specialized deposition for superconducting circuits (e.g., Niobium or Aluminum) and silicon-based qubit structures. To prevent 3D-printing outgassing from fouling sensitive qubits, we employ directional molecular shielding and localized cryo-pumping to "freeze" exhaust gases before they reach the operational QPU core, maintaining an ultra-pure fabrication void.

Essential non-printable components, such as high-precision control lasers and microwave pulse generators, will serve as the "printing bed." To maintain the integrity of nano-fabricated structures, we utilize In-Situ Integration. The printing assembly is housed within a pre-launched "mainframe" satellite. Once a QPU is "grown," robotic actuators transfer the chip from the printing vacuum into the operational cooling chamber without exposing it to atmospheric contaminants or gravitational stress. For expanded capacity, we utilize ISAM (In-Space Servicing, Assembly, and Manufacturing) robots to dock modular QPU arrays into a central orbiting lattice.

Noise Source	Terrestrial Impact	Orbital Impact	Estimated Reduction Factor
Seismic/Vibrational	High (requires heavy damping)	Near-zero (vacuum isolation)	10x - 100x
Gravitational Gradient	Perturbs delicate ion traps	Uniform microgravity	Significant (High-fidelity)
Thermal/Atmospheric	Limited by cryostat vacuum	Natural high-quality vacuum	10^{-10} Torr baseline

Orbital Selection and Stability
We prioritize Sun-Synchronous Orbit (SSO) over LEO to ensure thermal stability. Constant solar exposure prevents the expansion/contraction cycles found in LEO's 90-minute shadow crossings, which would otherwise introduce mechanical noise into the quantum state. Additionally, in an SSO, we bypass the diurnal cycle, providing 24/7 power for the manufacturing-compute duty cycle via constant solar arrays.

Radiation Mitigation and Edge Intelligence
Cosmic rays can induce "bit flips" and noise. Our mitigation strategy combines high-Z material shielding with Real-Time AI Error Correction. These proprietary models run on localized FPGA (Field Programmable Gate Array) hardware to eliminate bus latency, dynamically tuning microwave pulse sequences to compensate for ionizing radiation spikes at the hardware level.

Integrating AI into the fabrication phase allows for Closed-Loop Manufacturing. As the nanoprinter deposits material, AI sensors detect atomic-scale irregularities caused by solar flares or micrometeoroid vibrations, adjusting the deposition rate in real-time. This ensures that every orbital-grown QPU is optimized for its specific environmental variables, achieving a level of coherence and gate fidelity unreachable on Earth.

Connectivity and Scalability

Given the high-bandwidth requirements of quantum-classical hybrid models, we utilize Optical Communications (Lasercom). This provides the necessary throughput to link orbital compute nodes to Earth-based end-users with minimal latency. Initial "ink" (high-purity superconductors and substrate materials) will be launched in concentrated canisters via high-cadence heavy-lift vehicles. Long-term scalability targets the extraction of high-purity silicates from lunar regolith to minimize Earth-to-orbit launch costs and establish a fully extra-terrestrial supply chain.

Metric	Target Specification	Source/Rationale
Orbit	Sun-Synchronous (SSO), 500-600km	NASA ISAM Guidelines: Ensures constant solar power for the 3D printing thermal load.
Vacuum Level	10^{-10} Torr	Standard for Molecular Beam Epitaxy (MBE); space provides this "for free" in the wake of the satellite.
Launch Cost	$1,500/kg (Targeting SpaceX Starship)	CSIS Aerospace Security: Essential for proving the "NewSpace" economic model.
Thermal Delta	Operates at < 20 mK (milliKelvin)	Standard for superconducting qubits (IBM/Google benchmarks); requires active dilution refrigeration.
Power Budget	2.5 kW (Printing) / 5 kW (Computing)	Based on current SmallSat power capacities and cryocooler requirements.

Section 2: Potential Customers

Classical computing is approaching a hard ceiling. Binary, sequential processing fails catastrophically at combinatorial problems — molecular simulation, structural optimization, probabilistic inference — where the solution space scales exponentially with complexity. Quantum computing resolves this not by doing the same thing faster, but by doing something fundamentally different: representing and computing across all possible states simultaneously.

The industries that need this capability most urgently are those where computation directly gates scientific or economic progress.

Pharmaceuticals face the most consequential bottleneck. Modeling molecular interactions for a single drug candidate can require months of supercomputing time, even parallelized. For rare disease research, that delay has a direct humanitarian cost. Quantum simulation collapses this timeline from years to weeks by representing molecular configurations as quantum superpositions rather than sequential approximations. The global drug discovery market exceeded $68 billion in 2024, and the willingness to pay for faster, higher-fidelity simulation is well established.

Aerospace engineering and defense face a parallel constraint. High-fidelity fluid dynamics, structural stress, and materials simulations consume thousands of CPU hours per design iteration. Quantum optimization algorithms can search large parameter spaces orders of magnitude faster than classical heuristics — and for defense customers, quantum-accelerated simulation of hypersonic regimes and advanced materials represents sovereign capability no nation can afford to cede. This alignment of need and budget makes government and defense Qubital's natural first market.

AI and machine learning represent the largest eventual opportunity. Quantum processors are architecturally well-suited to probabilistic inference — the core operation of modern AI — and frontier AI labs and hyperscalers are already investing in quantum hardware access. Other high-complexity domains including financial modeling, logistics optimization, and climate simulation follow the same pattern: massive solution spaces, classical limits, and clear willingness to pay for superior compute. The priority order is defense first, pharma second, AI and enterprise cloud third — and Qubital's go-to-market is sequenced accordingly.

Section 3: Current Competition

The quantum computing industry in 2026 is crowded at the incumbent level and thinly populated at the space-based frontier. A competitive analysis reveals three distinct categories of players - each occupying a different position along the Earth-to-orbit axis - with Qubital positioned in the one category no competitor currently fills.

Terrestrial incumbents dominate by capital and scale. IBM, Google, IonQ (NYSE: IONQ, approximately $1.6 billion in cash reserves following its 2025 acquisitions of Capella Space and Qubitekk), Quantinuum, PsiQuantum, Atom Computing, and Pasqal are all pursuing Earth-based fabrication and Earth-based operation. These companies have collectively raised over $10 billion and dominate current quantum-as-a-service cloud offerings through partnerships with AWS Braket, Azure Quantum, and Google Cloud. Their shared limitation is dependence on cleanroom fabrication under Earth's gravitational and atmospheric constraints - the very constraints Qubital bypasses.

Space operators are the most direct strategic competitors. Infleqtion (NYSE: INFQ) is the category leader, having completed a SPAC-based IPO in February 2026 that raised over $550 million in gross proceeds. Infleqtion has flown neutral-atom quantum hardware on the International Space Station since 2018 through NASA's Cold Atom Laboratory and holds contracts exceeding $20 million with NASA JPL for the Quantum Gravity Gradiometer Pathfinder mission. IonQ is pursuing a comparable trajectory through its Capella Space acquisition, focused on satellite-based quantum networking. Rotonium, an Italian deep-tech startup with €1 million in seed funding, is developing room-temperature photonic quantum computers intended for eventual space deployment. Critically, all three space operators ship Earth-built hardware into orbit. None fabricate in space.

Orbital fabrication experimenters - Space Forge, United Semiconductors, and Redwire - have demonstrated in-orbit semiconductor crystal growth and thin-film manufacturing, with peer-reviewed results in *npj Microgravity* (2024) validating the physical premise that microgravity produces semiconductor crystals with fewer defects, larger volumes, and superior uniformity. None of these companies target quantum processors. Their work validates our scientific foundation without competing for our market.

Qubital is the only player combining both capabilities. Our competitive moat is the integration of in-orbit fabrication with space-native quantum operation - a technical bridge neither the quantum incumbents nor the orbital fabrication startups have attempted to build. This positioning is not a subdivision of the existing market; it is the creation of a new category.

Section 4: Addressable Markets

Qubital addresses a market opportunity defined by three compounding segments: space-based quantum computing, the broader quantum computing market, and quantum-as-a-service.

The space-based quantum computing segment is Qubital's beachhead. It was valued at approximately $1.55 billion in 2026 and is projected to reach $3.19 billion by 2030, representing roughly 19.9% annual growth. Growth is driven by defense and intelligence procurement, NASA mission awards, sovereign computer initiatives, and national-security demand. This is the segment where Qubital's differentiation is strongest because the company is built specifically around space-native fabrication and operation.

The broader quantum computing market is Qubital's adjacent opportunity. It grew from approximately $4.39 billion in 2025 to $5.59 billion in 2026 and is projected to exceed $25.63 billion by 2032. This market includes hardware, software, cloud access, and services across different qubit modalities and deployment environments. As the industry matures, demand for higher-fidelity quantum substrates and specialized quantum infrastructure will grow. Qubital can serve this demand by offering processors and compute capacity produced in an environment that Earth cannot replicate.

Quantum-as-a-service is the scalable revenue layer. This market was projected to grow from $2.3 billion in 2023 to $48.3 billion by 2033. The logic resembles the early cloud-computing and GPU-as-a-service markets: customers do not need to own the infrastructure; they pay for access to specialized compute when needed. This model fits Qubital especially well because orbital quantum capacity will be scarce, expensive, and technically complex to operate. Customers can access the benefit without owning or launching the hardware themselves.

Qubital's market entry strategy targets the beachhead first. The immediate serviceable market consists of government programs, defense primes, and national laboratories that need secure, sovereign, high-fidelity quantum compute. After technical validation, Qubital can expand into premium quantum-as-a-service for enterprise customers in pharmaceuticals, finance, logistics, AI, and materials science. The company does not need to win the entire quantum market. It needs to dominate the first high-value wedge, then expand into adjacent customers as capacity and credibility grow.

Section 5: Business Model and Revenue Strategy

Qubital's business model converts government-funded validation into recurring orbital compute revenue. Because the company is building deep-tech space infrastructure, it cannot rely on a conventional software launch model. Instead, Qubital monetizes in stages: non-dilutive R&D, paid pilots, reserved capacity, and usage-based quantum-as-a-service.

The first revenue stream is non-dilutive R&D. Qubital will pursue NASA InSPA, SBIR/STTR, DARPA, and other government research contracts to fund early technical proof without giving up equity. This aligns with the company's beachhead market because government and defense customers already have budgets for sovereign compute, quantum research, and space-based technology development. Non-dilutive funding reduces early financing pressure while creating validation from credible institutions.

The second revenue stream is paid proof-of-capability pilots. These pilots would be limited, milestone-based projects with defense primes, national laboratories, government agencies, or strategic partners. A pilot might test whether Qubital can fabricate quantum processor substrates in microgravity, whether substrate performance meets target thresholds, or whether a customer can integrate Qubital's compute into an existing simulation workflow. These are not generic demonstrations; they are paid validation programs with defined success metrics.

The third revenue stream is reserved quantum capacity. Once Qubital has orbital compute capability, high-priority customers can pay in advance for guaranteed or priority access. This is especially relevant for government labs, defense primes, and strategic hyperscalers that cannot rely on open-access, shared, or insecure infrastructure for mission-critical workloads. Reserved capacity creates recurring revenue before Qubital reaches mass commercial scale.

The fourth revenue stream is Qubital Cloud, a usage-based quantum-as-a-service platform. Enterprise customers in pharmaceuticals, materials science, finance, logistics, and AI would access orbital quantum compute through cloud or API-based interfaces. Pricing could be based on workload, compute time, reserved usage tiers, or specialized simulation packages. Over time, this creates the most scalable part of the business model.

Qubital sells three commercial products that map to these revenue streams. The first is the Orbital Fab Pilot: a paid proof-of-capability mission for NASA, DARPA, national labs, and defense primes. The second is Secure Quantum Capacity: dedicated access to sovereign, space-native quantum compute for mission-critical customers. The third is Qubital Cloud: a cloud-based quantum-as-a-service product for simulation-heavy enterprise workloads.

The full business model depends on a coordinated canvas. Qubital's customer segments are government labs, defense primes, strategic hyperscalers, and later commercial customers in pharma, finance, logistics, materials, and AI. Its value proposition is space-grown quantum processors, secure sovereign compute, higher-fidelity potential, faster simulation cycles, and lower environmental footprint. Its channels include NASA InSPA, SBIR/STTR, DARPA, defense-prime partnerships, strategic cloud partners, and enterprise sales. Its key partners are NASA or ISS National Lab, launch providers, payload integrators, university labs, defense primes, and cloud platforms. Its key activities are in-orbit nanoprinting, payload integration, quantum operations, AI compensation, customer integration, and IP protection. Its main costs are hardware R&D, payload development, launch integration, talent, mission assurance, IP/legal, and cloud infrastructure.

This model is expensive to build but hard to copy. If Qubital validates the platform, the company can move from a single space mission into a repeatable quantum infrastructure business.

Section 6: Our Ask

Qubital is raising a $5 million seed round to fund a 24-month operating plan with four specific, verifiable milestones that position the company for a Series A.

Use of funds: The round will be deployed across four priorities. Approximately $2.1 million (42%) will fund hardware research and development, including a ground-based prototype of the 3D nanoprinting process for quantum processor substrates, conducted in partnership with a university laser-thermal laboratory. Approximately $1.2 million (24%) will fund payload development and launch integration for an International Space Station National Lab mission, executed through NASA's In-Space Production Applications program. Approximately $1.1 million (22%) will fund a founding team of eight: two quantum hardware engineers, two aerospace engineers, one machine learning engineer, one fabrication specialist, one business development lead, and one government contracts lead. The remaining $600,000 (12%) covers legal counsel, intellectual property protection, cloud infrastructure, and working capital.

Milestones: First, a working ground prototype of the in-orbit nanoprinting process, validated by peer-reviewed publication within twelve months. Second, three signed letters of intent from enterprise customers - prioritizing pharmaceutical R&D leaders, defense primes, and national laboratories - establishing documented willingness to pay. Third, a manifested ISS payload slot via NASA InSPA with a confirmed launch window within the 24-month runway. Fourth, a first government contract award, targeting an SBIR Phase II or DARPA small-team contract in the $750,000 to $2 million range, establishing non-dilutive revenue.

Benchmarking: These milestones are calibrated against standard deep-tech Series A requirements: demonstrated technology readiness, customer validation, regulatory and partnership traction, and early revenue. Comparable seed rounds in the space-quantum category provide context - Rotonium raised €1 million in 2024 for a similar vision with a team of eight, while Infleqtion's pre-IPO capital stack extended into hundreds of millions. Our $5 million ask is positioned between these reference points, reflecting both the capital intensity of deep-tech space hardware and the disciplined scope of a 24-month runway targeting four concrete de-risking events.

We are seeking investors who understand that the next breakthrough in qubit fidelity will not come from a cleaner cleanroom on Earth - it will come from an environment Earth physically cannot replicate.

Section 7A: Validation Plan

Qubital's business model rests on four core assumptions, each of which must be tested within the seed runway before capital is committed to orbital development. Each hypothesis below has a defined methodology and a verifiable success criterion.

I. Technical: In-orbit nanoprinting produces viable quantum substrates. Peer-reviewed work in npj Microgravity (2024) confirms that microgravity produces semiconductor crystals with fewer defects and superior uniformity — but this has not been demonstrated for quantum processor substrates specifically. We will build a ground-based nanoprinting prototype in partnership with a university laser-thermal lab within twelve months, measure qubit coherence time and defect density against commercial terrestrial benchmarks, and submit results for peer review. Success criterion: Substrate coherence within 20% of terrestrial benchmarks; peer-reviewed manuscript submitted by Month 12.

II. Market: Government and defense customers will pay for orbital quantum computers. Starting Month 1, the business development lead will conduct structured discovery interviews across 20 organizations — defense primes, national laboratories, and government agencies — to assess willingness-to-pay and interest in codevelopment pilots. Success criterion: Three signed letters of intent from qualified organizations by Month 12.

III. Access: An ISS payload slot via NASA InSPA is achievable within 24 months. We will engage the ISS National Lab and at least two payload integrators (Starlab Space, Axiom Space) beginning Month 2 and submit a formal InSPA application by Month 6, incorporating ground prototype results. Success criterion: Manifested ISS payload slot with confirmed launch window by Month 18.

IV. Revenue: Non-dilutive government contracts are accessible before Series A. The government contracts lead will identify and submit to at least four active SBIR, STTR, or DARPA solicitations within the first twelve months, targeting DARPA's Quantum Benchmarking Initiative and AFRL Quantum Information Science programs. Success criterion: At least one contract award of $750K–$2M (SBIR Phase II or DARPA small-team) by Month 24.

Section 7B: Milestones

The $5M seed round funds a 24-month plan structured as four phases, each targeting a specific category of risk reduction. The four milestones delivered — peer-reviewed prototype, three LOIs, manifested ISS payload, and first government contract — collectively satisfy the technology readiness, customer validation, regulatory traction, and early revenue pillars required for a credible deep-tech Series A.

Phase 1: Foundation (Months 0–6)

Founding team of eight hired and onboarded by Month 3. Provisional patent applications filed for the nanoprinting process and AI compensation stack by Month 4. University quantum lab partnerships signed by Month 5. First SBIR Phase I applications submitted by Month 6.

Phase 2: Proof (Months 6–12)

Ground nanoprinting prototype operational and producing measurable substrate samples by Month 9. Substrate performance benchmarked against terrestrial quantum processors by Month 11. Peer-reviewed manuscript submitted by Month 12. Three signed LOIs from qualified government or enterprise customers by Month 12.

Phase 3: Access (Months 12–18)

NASA InSPA payload application submitted by Month 14, informed by ground prototype data. Active DARPA program engagement initiated by Month 15. First government contract awarded — SBIR Phase II or DARPA small-team, $750K–$2M — by Month 18. ISS payload slot manifested with confirmed launch window by Month 18.

Phase 4: Orbit

(Months 18–24) Peer-reviewed publication accepted by Month 20. Series A fundraising materials finalized and investor outreach launched by Month 21. First ISS payload integration test executed by Month 23 if the launch window falls within the runway. Series A term sheet signed by Month 24. These milestones are calibrated against Infleqtion's pre-IPO de-risking trajectory and Rotonium's comparable €1M seed scope. The $5M ask reflects the capital intensity of deep-tech space hardware while maintaining a disciplined 24-month runway with four individually verifiable output

References:

Kaltenbaek, R., Acin, A., Bacsardi, L., Bianco, P., Bouyer, P., Diamanti, E., Marquardt, C., Omar, Y., Pruneri, V., Rasel, E., Sang, B., Seidel, S., Ulbricht, H., Ursin, R., Villoresi, P., Bossche, M. v. d., von Klitzing, W., Zbinden, H., Paternostro, M., & Bassi, A. (2021). Quantum Technologies in Space. *Exp Astron (2021)*. https://doi.org/10.1007/s10686-021-09731-x

Market.us. (2024). *Quantum computing-as-a-service (QCaaS) market size, share | CAGR of 35.6%*. https://market.us/report/quantum-computing-as-a-service-qcaas-market/

Research and Markets. (2026). *Quantum computing market - Global forecast 2026-2032*. https://www.researchandmarkets.com/reports/5470718/quantum-computing-market-global-forecast-2026

Research and Markets. (2026). *Space-based quantum computing market report 2026*. https://www.researchandmarkets.com/reports/6167110/space-based-quantum-computing-market-report